

Aptitude Masterclass: Time, Work & Pipes

Comprehensive Guide and 100 MCQ Practice Set

Quantitative Aptitude Series

1 Theoretical Mastery Guide

1.1 The Fundamental Axiom

The entirety of Time and Work logic is built upon a single proportional relationship:

$$\text{Total Work} = \text{Efficiency} \times \text{Time}$$

Where:

- **Work:** Usually assumed as 1 unit (Unitary Method) or the **LCM** of all given time periods to avoid fractions.
- **Efficiency:** The work done per unit time. For a fixed work, Efficiency is inversely proportional to Time: $E \propto \frac{1}{T}$.

1.2 Core Formulas & Shortcuts

1. **Two Persons Together:** If A takes x days and B takes y days:

$$\text{Time taken together} = \frac{xy}{x + y}$$

2. **The MDH Chain Rule:** For comparing two different groups:

$$\frac{M_1 D_1 H_1 E_1}{W_1} = \frac{M_2 D_2 H_2 E_2}{W_2}$$

(M = Men, D = Days, H = Hours, E = Efficiency, W = Work/Wages)

3. **Pipes and Cisterns:**

- Inlet Pipe (Filling) = Positive Efficiency.
- Outlet Pipe/Leak (Emptying) = Negative Efficiency.
- If a tank is filled in x hours and emptied in y hours, net work in 1 hour = $\frac{1}{x} - \frac{1}{y}$.

1.3 Edge Cases & Nuances

- **Alternating Days:** If A and B work on alternate days, find the work done in a 2-day cycle. Divide total work by cycle work to find full cycles, then calculate remaining work manually.
- **Efficiency Ratios:** "A is 3 times more efficient than B" means $E_A = 4E_B$. "A is 3 times as efficient as B" means $E_A = 3E_B$.
- **Negative Work (Pipes):** If the emptying rate is higher than the filling rate, the tank will never fill.

2 Practice Set (Questions 1–100)

1. A can do a piece of work in 12 days and B can do it in 15 days. They work together for 5 days and then B leaves. In how many days will A finish the remaining work?
(A) 3 days
(B) 5 days
(C) 10 days
(D) 8 days
2. X, Y and Z can complete a work in 10, 20 and 30 days respectively. Working together, in how many days can they complete the work?
(A) 6 days
(B) $5\frac{5}{11}$ days
(C) 7 days
(D) 11 days
3. A can do a work in 15 days and B in 20 days. If they work on it together for 4 days, then the fraction of the work that is left is:
(A) $\frac{1}{4}$
(B) $\frac{1}{10}$
(C) $\frac{7}{15}$
(D) $\frac{8}{15}$
4. A is thrice as efficient as B. If A takes 60 days less than B to complete a work, find the time in which they can do it together.
(A) 22.5 days
(B) 30 days
(C) 25 days
(D) 45 days
5. If 12 men and 16 boys can do a piece of work in 5 days; 13 men and 24 boys can do it in 4 days, then the ratio of the daily work done by a man to that of a boy is:
(A) 2:1
(B) 3:1
(C) 3:2
(D) 6:7
6. A is 50% more efficient than B. If B takes 21 days to complete a work, how many days will A take?
(A) 14 days
(B) 10.5 days
(C) 18 days
(D) 12 days

7. A is 200% more efficient than B. If they together finish a work in 12 days, how many days will B take alone?
- (A) 36 days
 - (B) 48 days
 - (C) 24 days
 - (D) 60 days
8. A and B can do a work in 12 days. B and C in 15 days, C and A in 20 days. If A, B and C work together, they will complete the work in:
- (A) 5 days
 - (B) 10 days
 - (C) 12 days
 - (D) 15 days
9. In the above question, how many days will A alone take?
- (A) 30 days
 - (B) 40 days
 - (C) 20 days
 - (D) 60 days
10. A is twice as fast as B and B is thrice as fast as C. The journey covered by C in 42 minutes will be covered by A in:
- (A) 7 min
 - (B) 14 min
 - (C) 28 min
 - (D) 63 min
11. A can finish a work in 18 days and B can do the same work in half the time taken by A. Then, working together, what part of the same work they can finish in a day?
- (A) $\frac{1}{6}$
 - (B) $\frac{1}{9}$
 - (C) $\frac{2}{5}$
 - (D) $\frac{2}{7}$
12. A, B and C can complete a piece of work in 24, 6 and 12 days respectively. Working together, they will complete the same work in:
- (A) $\frac{1}{24}$ day
 - (B) $\frac{7}{24}$ day
 - (C) $3\frac{3}{7}$ days
 - (D) 4 days
13. A can do a certain work in the same time in which B and C together can do it. If A and B together could do it in 10 days and C alone in 50 days, then B alone could do it in:
- (A) 15 days

- (B) 20 days
(C) 25 days
(D) 30 days
14. A and B can decide to do a piece of work in 45 days. A alone can do it in 60 days. How long will B take to do it?
- (A) 180 days
(B) 120 days
(C) 90 days
(D) 150 days
15. A can do a work in 4 days, B in 5 days and C in 10 days. Find the time taken by A, B and C together.
- (A) 2 days
(B) $1\frac{9}{11}$ days
(C) 3 days
(D) 4 days
16. 10 men can complete a work in 7 days. But 10 women need 14 days. If 5 men and 10 women work together, how many days will they take?
- (A) 5 days
(B) 7 days
(C) 6 days
(D) 8 days
17. A, B and C together can earn Rs. 300 per day, while A and C together earn Rs. 188 and B and C together earn Rs. 152. The daily earning of C is:
- (A) Rs. 40
(B) Rs. 68
(C) Rs. 112
(D) Rs. 150
18. A is 30% more efficient than B. How much time will they, working together, take to complete a job which A alone could have done in 23 days?
- (A) 13 days
(B) 11 days
(C) 20.3 days
(D) None
19. 2 men and 3 boys can do a piece of work in 10 days while 3 men and 2 boys can do the same work in 8 days. In how many days can 2 men and 1 boy do the same work?
- (A) 12.5 days
(B) 15 days
(C) 10 days

- (D) 8 days
20. If 3 men or 6 women can do a piece of work in 16 days, in how many days can 12 men and 8 women do the same work?
- (A) 3 days
(B) 4 days
(C) 5 days
(D) 2 days
21. A and B can do a piece of work in 72 days; B and C in 120 days; A and C in 90 days. In what time can A alone do it?
- (A) 120 days
(B) 80 days
(C) 100 days
(D) 150 days
22. 4 men and 6 women can complete a work in 8 days, while 3 men and 7 women can complete it in 10 days. In how many days will 10 women complete it?
- (A) 35 days
(B) 40 days
(C) 45 days
(D) 50 days
23. A takes twice as much time as B or thrice as much time as C to finish a piece of work. Working together, they can finish the work in 2 days. B can do the work alone in:
- (A) 4 days
(B) 6 days
(C) 8 days
(D) 12 days
24. A can do a certain work in 12 days. B is 60% more efficient than A. How many days will B and A together take to finish the same work?
- (A) $4\frac{8}{13}$ days
(B) $3\frac{3}{5}$ days
(C) $5\frac{2}{3}$ days
(D) 6 days
25. 20 men can prepare 40 toys in 30 days working 5 hours a day. In how many days can 12 men prepare 24 toys working 8 hours a day?
- (A) 18.75 days
(B) 20 days
(C) 25 days
(D) 15 days
26. A and B can do a work in 18 and 24 days respectively. They worked together for 8 days and then A left. The remaining work was finished by B in:

- (A) 5 days
 - (B) $5\frac{1}{3}$ days
 - (C) 8 days
 - (D) 10 days
27. A, B and C can do a work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?
- (A) 12 days
 - (B) 15 days
 - (C) 16 days
 - (D) 18 days
28. A is twice as good a workman as B and together they finish a piece of work in 18 days. In how many days will A alone finish the work?
- (A) 27 days
 - (B) 25 days
 - (C) 24 days
 - (D) 30 days
29. A can do a piece of work in 80 days. He works at it for 10 days and then B alone finishes the remaining work in 42 days. In how much time will A and B working together finish the work?
- (A) 25 days
 - (B) 30 days
 - (C) 35 days
 - (D) 40 days
30. 45 men can complete a work in 16 days. Six days after they started working, 30 more men joined them. How many days will they now take to complete the remaining work?
- (A) 6 days
 - (B) 8 days
 - (C) 5 days
 - (D) 4 days
31. A, B and C can do a piece of work in 11 days, 20 days and 55 days respectively, working alone. How soon can the work be done if A is assisted by B and C on alternate days?
- (A) 7 days
 - (B) 8 days
 - (C) 9 days
 - (D) 10 days
32. A and B together can do a piece of work in 30 days. A having worked for 16 days, B finishes the remaining work alone in 44 days. In how many days shall B finish the whole work alone?
- (A) 30 days

- (B) 40 days
(C) 60 days
(D) 70 days
33. If 5 men or 9 women can do a piece of work in 19 days, then in how many days will 3 men and 6 women do the same work?
- (A) 12 days
(B) 15 days
(C) 18 days
(D) 21 days
34. A can do a work in 10 days and B in 20 days. They work together for 5 days and then A goes away. In how many days will B finish the work?
- (A) 5 days
(B) 6 days
(C) 8 days
(D) 10 days
35. 10 men and 15 women together can complete a work in 6 days. It takes 100 days for one man alone to complete the same work. How many days will be required for one woman alone to complete the same work?
- (A) 90 days
(B) 145 days
(C) 150 days
(D) 225 days
36. A and B can do a piece of work in 30 days, while B and C can do the same work in 24 days and C and A in 20 days. They all work together for 10 days when B and C leave. How many days more will A take to finish the work?
- (A) 18 days
(B) 24 days
(C) 30 days
(D) 36 days
37. Seven men can complete a work in 12 days. They started the work and after 5 days, two men left. In how many days will the work be completed by the remaining men?
- (A) 9.8 days
(B) 8 days
(C) 10 days
(D) 12 days
38. A can do a piece of work in 10 days, B in 15 days. They work for 5 days. The rest of the work was finished by C in 2 days. If they get Rs. 1500 for the whole work, the daily wages of B and C together are:
- (A) Rs. 150

- (B) Rs. 225
(C) Rs. 250
(D) Rs. 300
39. A, B and C can do a piece of work in 36, 54 and 72 days respectively. They started the work but A left 8 days before the completion of the work while B left 12 days before the completion. The number of days for which C worked is:
- (A) 24 days
(B) 28 days
(C) 32 days
(D) 36 days
40. If 3 men and 5 women do a piece of work in 8 days and 2 men and 7 children do the same in 12 days. How many women can do as much work in a day as 21 children?
- (A) 10
(B) 15
(C) 12
(D) 18
41. A pipe can fill a tank in 15 hours. Due to a leak in the bottom, it is filled in 20 hours. If the tank is full, how much time will the leak take to empty it?
- (A) 60 hrs
(B) 50 hrs
(C) 45 hrs
(D) 30 hrs
42. Pipe A can fill a tank in 20 min and Pipe B in 30 min. If both are opened together, the time taken to fill the tank is:
- (A) 12 min
(B) 15 min
(C) 25 min
(D) 50 min
43. A tap can fill a tank in 6 hours. After half the tank is filled, three more similar taps are opened. The total time taken to fill the tank completely is:
- (A) 3 hrs 15 min
(B) 3 hrs 45 min
(C) 4 hrs
(D) 4 hrs 15 min
44. Three pipes A, B and C can fill a tank in 30 min, 20 min and 10 min. If all are opened together, what is the proportion of liquid from pipe C in the tank after 3 minutes?
- (A) $\frac{5}{11}$
(B) $\frac{6}{11}$
(C) $\frac{7}{11}$

- (D) $\frac{8}{11}$
45. Two pipes A and B can fill a tank in 24 min and 32 min. After how much time should B be closed so the tank is full in 18 minutes?
- (A) 5 min
(B) 6 min
(C) 8 min
(D) 10 min
46. Two pipes fill a tank in 20 and 24 min. A waste pipe empties 3 gal/min. Together they fill in 15 min. Tank capacity?
- (A) 120 gal
(B) 60 gal
(C) 100 gal
(D) 150 gal
47. Pipe C is twice as fast as B, B is twice as fast as A. Together they fill in 5 hours. Pipe A alone takes:
- (A) 35 hrs
(B) 25 hrs
(C) 20 hrs
(D) 40 hrs
48. A and B together fill in 4 hrs. If B takes 6 hrs more than A separately, how long does A take?
- (A) 2 hrs
(B) 6 hrs
(C) 8 hrs
(D) 4 hrs
49. One pipe fills 3 times as fast as another. Together they fill in 36 minutes. The slower pipe alone takes:
- (A) 108 min
(B) 144 min
(C) 81 min
(D) 192 min
50. Second pipe fills 5 hrs faster than first and 4 hrs slower than third. First two pipes together = third alone. Time for first?
- (A) 15 hrs
(B) 10 hrs
(C) 6 hrs
(D) 30 hrs
51. A fills in 12 min, B in 15 min. A is closed after 3 minutes. How much *more* time for B?

- (A) 8 min 15 sec
(B) 7 min
(C) 10 min
(D) 12 min
52. Normal fill is 8 hrs. Leak makes it 10 hrs. If full, the leak empties it in:
(A) 40 hrs
(B) 20 hrs
(C) 50 hrs
(D) 60 hrs
53. A in 15 min, B in 20 min. A is turned off at 4 minutes. Total time?
(A) 14 min 40 sec
(B) 11 min 40 sec
(C) 12 min 30 sec
(D) 10 min 20 sec
54. A, B, C fill in 6 hrs. After 2 hrs, C is closed. A and B fill the rest in 7 hrs. C alone takes:
(A) 14 hrs
(B) 12 hrs
(C) 10 hrs
(D) 16 hrs
55. Obstruction reduces A flow to $\frac{7}{8}$ and B to $\frac{5}{6}$. Once removed, cistern fills in 3 min. Clogging duration?
(A) 4.5 min
(B) 2.5 min
(C) 3.5 min
(D) 5.5 min
56. A in 12 min, B in 15 min, C empties in 6 min. A and B open for 5 min, then C also opened. Empty time?
(A) 45 min
(B) 33 min
(C) 30 min
(D) 60 min
57. 12 buckets (13.5L) fill a tank. How many buckets needed if capacity is 9L?
(A) 18
(B) 15
(C) 16
(D) 8
58. A fills in 16 min, B empties in 8 min. If half full and both open, tank will be:

- (A) emptied in 8 min
 - (B) filled in 12 min
 - (C) emptied in 12 min
 - (D) filled in 8 min
59. A in 37.5 min, B in 45 min. If full in 30 min, after how long was B turned off?
- (A) 9 min
 - (B) 5 min
 - (C) 10 min
 - (D) 15 min
60. C is twice as fast as B, B twice as fast as A. If C fills in 10 min, A takes:
- (A) 40 min
 - (B) 20 min
 - (C) 30 min
 - (D) 60 min
61. A fills in 10 hrs. Leak makes it 12 hrs. Leak empties in:
- (A) 60 hrs
 - (B) 30 hrs
 - (C) 40 hrs
 - (D) 24 hrs
62. A, B fill in 12, 16 hrs. C empties in 8 hrs. If all open, time taken?
- (A) 48 hrs
 - (B) 24 hrs
 - (C) 36 hrs
 - (D) 12 hrs
63. A fills in 4 hrs, B empties in 6 hrs. Both open, tank fills in:
- (A) 12 hrs
 - (B) 10 hrs
 - (C) 8 hrs
 - (D) 24 hrs
64. 3 pipes fill in 10, 20, 30 min. Together they fill in:
- (A) 5.45 min
 - (B) 6 min
 - (C) 7.5 min
 - (D) 10 min
65. A in 5h, B in 4h, C empties in 2h. If all open, the tank will be:
- (A) Emptied
 - (B) Filled

- (C) No change
 - (D) None
66. In the above question, total time for state change:
- (A) 20 hrs
 - (B) 10 hrs
 - (C) 15 hrs
 - (D) 25 hrs
67. 2 taps fill in 12 and 15 min. Outlet for 10 min. All open, time?
- (A) 20 min
 - (B) 12 min
 - (C) 15 min
 - (D) 10 min
68. A in 12m, B in 15m. C empties at 6 gal/min. All open = 20 min. Capacity?
- (A) 60 gal
 - (B) 120 gal
 - (C) 90 gal
 - (D) 45 gal
69. A fills in 8 hrs. After 4 hrs, another identical tap opens. Total time?
- (A) 6 hrs
 - (B) 5 hrs
 - (C) 7 hrs
 - (D) 8 hrs
70. A in 10h, B in 12h, C in 15h. A constant, B and C alternate hourly. Time?
- (A) 5 hrs
 - (B) 6 hrs
 - (C) 7 hrs
 - (D) 8 hrs
71. Leak empties in 4h. Tap adds 2L/min. Both open = empty in 6h. Capacity?
- (A) 1440 L
 - (B) 1200 L
 - (C) 720 L
 - (D) 2880 L
72. A in 3h, B in 4h. C empties in 2h. Open at 3, 4, 5 PM. Tank full at?
- (A) 7:12 PM
 - (B) 8:15 PM
 - (C) 9:00 PM
 - (D) 6:45 PM

73. Pipes 20, 24 min. Waste 3 gal/min. Together 15 min. Capacity?
- (A) 120 gal
 - (B) 60 gal
 - (C) 180 gal
 - (D) 240 gal
74. $\frac{2}{5}$ full. A fills in 10 min, B empties in 6 min. Result?
- (A) 6 min empty
 - (B) 6 min fill
 - (C) 15 min empty
 - (D) 15 min fill
75. A in 12m, B in 18m. After 2 min, A closed. More time for B?
- (A) 15 min
 - (B) 12 min
 - (C) 13 min
 - (D) 14 min
76. A in 20m, B in 25m. Both for 5 min, then B closed. Total time?
- (A) 16 min
 - (B) 17 min
 - (C) 15 min
 - (D) 18 min
77. A in 12m, B in 15m. A closed 3 min before B fills. Total time?
- (A) 8 min
 - (B) 7 min
 - (C) 9 min
 - (D) 10 min
78. A in 10m, B in 12m. C empties 2 gal/min. All open = 6 min. Capacity?
- (A) 30 gal
 - (B) 40 gal
 - (C) 60 gal
 - (D) 20 gal
79. A in 10h. Leak makes it 12h. Leak empties full in:
- (A) 60 hrs
 - (B) 40 hrs
 - (C) 50 hrs
 - (D) 30 hrs
80. A is 4x faster than B. A fills in 15 min. B takes:

- (A) 60 min
 - (B) 45 min
 - (C) 30 min
 - (D) 75 min
81. A, B in 20, 30 min. A closed 5 min before full. Total time?
- (A) 14 min
 - (B) 12 min
 - (C) 15 min
 - (D) 16 min
82. A in 12h, B in 16h, C empties in 8h. All open. Time?
- (A) 48 hrs
 - (B) 24 hrs
 - (C) 36 hrs
 - (D) 12 hrs
83. 12, 18 min. A closed after 4 min. B takes:
- (A) 8 min
 - (B) 6 min
 - (C) 10 min
 - (D) 12 min
84. 3, 4 hrs fill, 1 hr empty. Open at 3, 4, 5 PM. Empty at?
- (A) 7:12 PM
 - (B) 7:15 PM
 - (C) 8:12 PM
 - (D) 6:15 PM
85. Tap in 6h. After 3h, 3 more taps. Total time?
- (A) 3 hr 45 min
 - (B) 4 hr
 - (C) 3 hr 15 min
 - (D) 4 hr 30 min
86. Fill in 8h. Leak makes it 10h. Leak empties in:
- (A) 40 hrs
 - (B) 20 hrs
 - (C) 30 hrs
 - (D) 50 hrs
87. A in 15m, B in 20m. B closed after 4 min. Total time?
- (A) 12 min
 - (B) 10 min

- (C) 15 min
 - (D) 14 min
88. Fill in 9h. Leak makes it 10h. Leak empties in:
- (A) 90 hrs
 - (B) 100 hrs
 - (C) 80 hrs
 - (D) 95 hrs
89. A in 10m, B in 15m. A closed after 2 min. Total time?
- (A) 12 min
 - (B) 10 min
 - (C) 14 min
 - (D) 15 min
90. A in 20m, B in 30m. B closed 5 min before full. Time?
- (A) 15 min
 - (B) 12 min
 - (C) 14 min
 - (D) 16 min
91. A in 12m, B in 15m. A closed after 3 min. Total time?
- (A) 11.25 min
 - (B) 10 min
 - (C) 12 min
 - (D) 13 min
92. 8h, 12h, 6h empty. All open. Time?
- (A) 24 hrs
 - (B) 12 hrs
 - (C) 8 hrs
 - (D) 16 hrs
93. 2h, 3h, 6h empty. All open. Time?
- (A) 1.5 hrs
 - (B) 1 hr
 - (C) 2 hrs
 - (D) 2.5 hrs
94. 5h, 10h, 20h empty. All open. Time?
- (A) 4 hrs
 - (B) 3.5 hrs
 - (C) 5 hrs
 - (D) 4.5 hrs

95. 15m, 10m, 5m empty. All open. Result?
- (A) 30 min empty
 - (B) 30 min fill
 - (C) 20 min empty
 - (D) 10 min fill
96. 4h, 6h. B closed after 2h. Total time?
- (A) 2.67 hrs
 - (B) 3 hrs
 - (C) 2.5 hrs
 - (D) 4 hrs
97. 12m, 18m. A closed after 4 min. B finishes in?
- (A) 8 min
 - (B) 6 min
 - (C) 10 min
 - (D) 12 min
98. Leak 8h. Tap 6L/min. Both open = 12h. Capacity?
- (A) 8640 L
 - (B) 4320 L
 - (C) 7200 L
 - (D) 5400 L
99. 10m, 20m alternate. Start A. Time?
- (A) 13 min 20 sec
 - (B) 15 min
 - (C) 14 min
 - (D) 12 min
100. 20m, 30m alternate. Start A. Time?
- (A) 24 min
 - (B) 25 min
 - (C) 26 min
 - (D) 22 min

Answer Key

1. A	21. A	41. A	61. A	81. A
2. B	22. B	42. A	62. A	82. A
3. D	23. B	43. B	63. A	83. A
4. A	24. A	44. B	64. A	84. A
5. A	25. A	45. C	65. B	85. A
6. A	26. B	46. A	66. A	86. A
7. B	27. B	47. A	67. A	87. A
8. B	28. A	48. A	68. B	88. A
9. A	29. B	49. C	69. A	89. A
10. A	30. A	50. B	70. A	90. A
11. A	31. B	51. B	71. A	91. A
12. C	32. C	52. A	72. A	92. A
13. C	33. B	53. A	73. A	93. A
14. A	34. A	54. C	74. A	94. A
15. B	35. C	55. C	75. A	95. A
16. B	36. A	56. A	76. A	96. A
17. B	37. A	57. D	77. A	97. A
18. A	38. A	58. D	78. A	98. A
19. A	39. C	59. B	79. A	99. A
20. A	40. C	60. A	80. A	100. A